

Deep Learning for Organoid Analysis: Graph-Based Modeling of 3D Cellular Architectures

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ANR Morpheus Project

Thesis Defense



Organoids: Mini-Organs in a Dish

What are organoids?

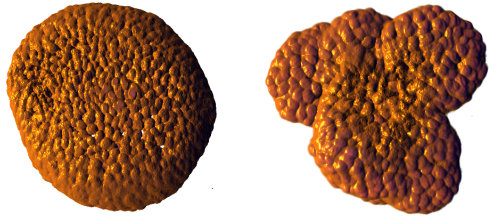
- 3D structures grown *in vitro* from stem cells
- Self-organize to mimic real organ architecture
- Reproduce spatial complexity observed *in vivo*

Key features:

- Form layers and internal cavities (lumens)
- Exhibit cellular polarization
- Bridge gap between 2D cultures and animal models

Applications:

- Personalized medicine
- Drug screening
- Disease modeling



Prostate Organoids: Two Phenotypes

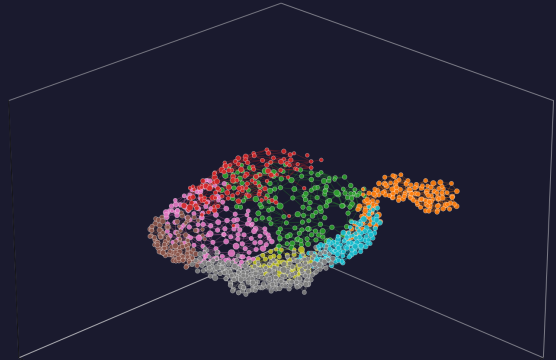
✓ Cystic (Healthy)

- Spherical, central cavity
- Organized structure
- **Uniform** cell distribution

⚠ Cauliflower (Perturbed)

- Irregular, multiple buds
- Disorganized structure
- **Clustered** cell distribution

Cauliflower Organoid - Cluster Analysis
(Hierarchical Clustering)



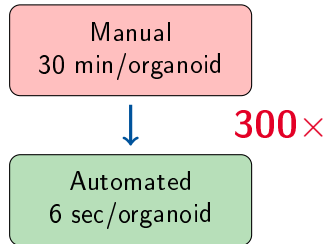
Challenge #1: Automatic Quantification

Manual analysis problems:

- **✗ Subjective** — inter-expert variability
- **✗ Inconsistent** — intra-expert variability
- **✗ Slow** — 30 min/organoid

Need:

- ✓ **Automated** — scalable
- ✓ **Objective** — reproducible
- ✓ **Fast** — high-throughput



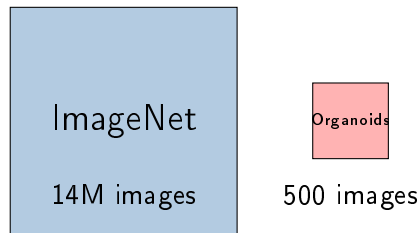
Challenge #2: Rare Annotated Data

Computer Vision:

- ImageNet: 14 million labeled images
- Public, freely available

Organoid Domain:

- No public datasets
- Expert annotation: $\sim 50\text{€}/\text{hour}$
- 1000s of annotations = impractical



Data Scarcity

Fundamental bottleneck for training deep learning models

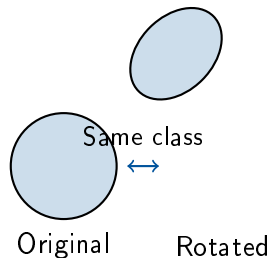
Challenge #3: Geometric Robustness

Organoids in 3D culture:

- No preferred orientation
- Random position and rotation

Requirements:

- **Invariance** to rotations
- **Invariance** to translations
- Robustness to acquisition variations



Key Insight

Same organoid, different orientation $\Rightarrow$ same prediction

Q1 Are geometric graphs effective for organoid representation?

- Can we capture biological information while compressing data?

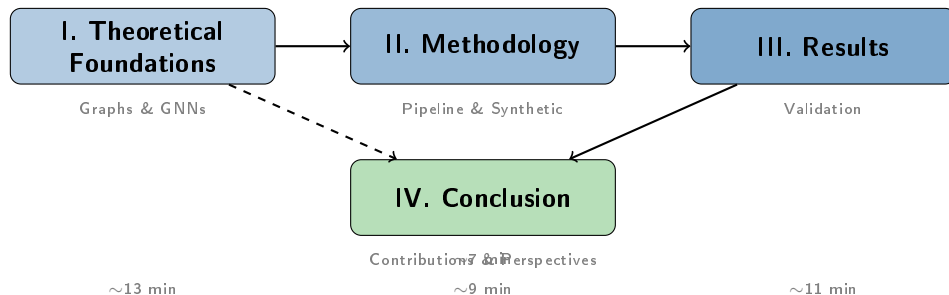
Q2 Do equivariant GNNs outperform classical approaches?

- Do they provide geometric robustness?

Q3 Is transfer learning from synthetic data effective?

- Can synthetic organoids compensate for lack of annotations?

Presentation Outline



Total: ~40 minutes + questions

Part I

Theoretical Foundations

Graphs and Graph Neural Networks

Why Not 3D CNNs?

Memory Constraints:

- Single organoid: ~ 2 GB
- $2048 \times 2048 \times 200$ voxels
- GPU memory: prohibitive

Downsampling Issues:

- Destroys cellular details
- Individual cells unresolvable

No Native Invariance:

- Not invariant to 3D rotations
- Requires extensive augmentation
- No guarantees

Grid-Based Limitation:

- Treats organoid as voxel grid
- Doesn't model cells as entities
- No relational structure

Conclusion

3D CNNs are not adapted to our constraints

Graph Representation: A Natural Abstraction

Key Insight:

Cells form a network of spatial interactions

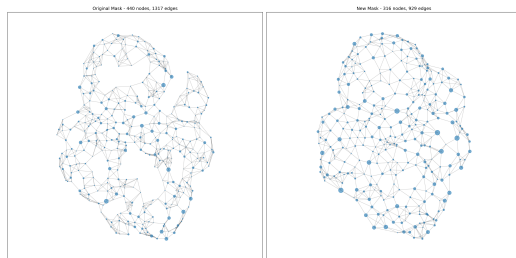
Our Abstraction:

- Each **cell** $\rightarrow$ **node**
- Spatial proximity $\rightarrow$ **edges**
- Node features: position, volume

Compression:

GB (images) $\rightarrow$ MB (graphs)

1000 $\times$ compression



Graph Neural Networks: Message Passing

Challenge: Graphs are non-Euclidean

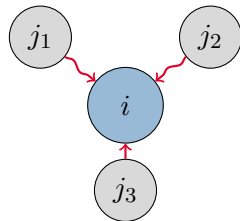
- Variable neighbors, no ordering

Solution: Message Passing

$$h_i^{(l+1)} = \text{UPD} \left(h_i^{(l)}, \text{AGG} \left(\{h_j^{(l)}\}_{j \in \mathcal{N}(i)} \right) \right)$$

Iterative process:

- 1 Each node sends message to neighbors
- 2 Aggregate incoming messages
- 3 Update node representation



Messages flow to central node

► Animation: `message_passing.gif`

Global Pooling and Classification

Goal: Graph-level prediction (not node-level)

Global Pooling:

- **Mean pooling:** $h_G = \frac{1}{|V|} \sum_{i \in V} h_i$
- **Max pooling:** $h_G = \max_{i \in V} h_i$
- **Attention pooling:** learned importance weights

Classification Head:

Node features $\xrightarrow{\text{GNN layers}}$ Node embeddings $\xrightarrow{\text{Pooling}}$ h_G $\xrightarrow{\text{MLP}}$ $\hat{y}$

GCN: Graph Convolutional Network (Baseline)

Kipf & Welling, 2017

Update Rule:

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i) \cup \{i\}} \frac{1}{\sqrt{d_i d_j}} W^{(l)} h_j^{(l)} \right)$$

Properties:

- Normalized mean aggregation
- Simple and effective
- Treats all neighbors equally
- No distance/importance weighting

GAT: Graph Attention Network (Main Choice)

Veličković et al., 2018

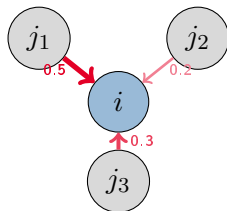
Key Idea: Learn attention coefficients to weight neighbors differently

1. **Compute attention scores:**

$$\alpha_{ij} = \text{softmax}_j (\text{score}(h_i, h_j))$$

2. **Weighted aggregation:**

$$h_i^{\text{new}} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} \cdot W h_j \right)$$



Learned attention weights

✓ **Best performance** — Multi-head attention for robustness

Zaheer et al., 2017

Approach: Treat organoid as unordered set of cells

$$f(X) = \rho \left(\sum_{x \in X} \phi(x) \right)$$

Properties:

- Permutation invariant
- Ignores local spatial structure
- Each cell processed independently before aggregation

Observation: Performs reasonably well

⇒ Global statistics are informative, but local structure helps

Geometric Equivariance: Key Concept

Why for organoids?

No preferred orientation in 3D culture

Invariance (classification)

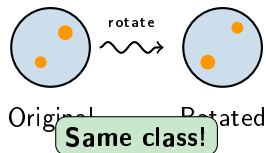
$$f(T(x)) = f(x)$$

Same output after rotation

Equivariance (features)

$$f(T(x)) = T(f(x))$$

Features rotate coherently



rotation_invariance.gif

The $E(3)$ Group: 3D Euclidean Symmetries

$E(3)$ = Euclidean group in 3 dimensions

Rotations

Any 3D rotation
around any axis

Translations

Shifting the entire
structure in space

Reflections

Mirror
symmetries

Key Insight

We can build architectures that **guarantee** these invariances by construction — not learned through augmentation, but mathematically guaranteed.

EGNN: Equivariant Graph Neural Network

Satorras et al., 2021

Key Principle: Use only invariant quantities in messages

- Distances are invariant (don't change under rotation)
- Direction vectors are equivariant

Practical Advantages:

- ✓ No data augmentation needed
- ✓ Better generalization with less data
- ✓ Robustness guaranteed by construction

Trade-off: Slightly lower raw performance, but guaranteed geometric robustness

EGNN: Key Formulas

Message computation (invariant):

$$m_{ij} = \phi_m(h_i, h_j, \|x_i - x_j\|^2)$$

Coordinate update (equivariant):

$$x_i^{(l+1)} = x_i^{(l)} + \sum_{j \neq i} (x_i - x_j) \cdot \phi_x(m_{ij})$$

Feature update:

$$h_i^{(l+1)} = \phi_h \left(h_i^{(l)}, \sum_{j \in \mathcal{N}(i)} m_{ij} \right)$$

Best of Both Worlds

Invariant messages + Equivariant coordinate updates

Architecture Comparison Summary

Architecture	Aggregation	E(3)-Equiv.	Strength
GCN	Mean	✗	Simple baseline
GAT	Attention	✗	Best accuracy
DeepSets	Global sum	✗	No local structure
EGNN	Distance-based	✓	Guaranteed robustness

Key Trade-off:

GAT → Best raw performance

EGNN → Guaranteed geometric invariance

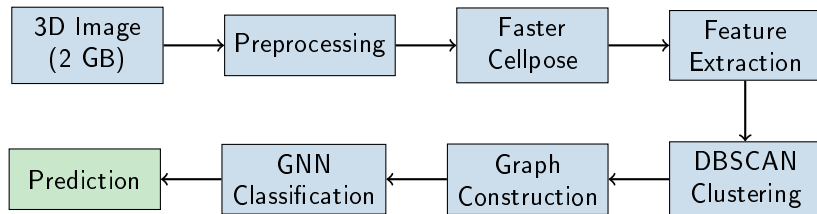
- ① **Graphs** capture relational structure with **1000× compression**
- ② **GNNs** learn from non-Euclidean data via message passing
- ③ **GAT** achieves best performance through adaptive attention
- ④ **EGNN** provides guaranteed geometric robustness

Part II

Methodology

From Raw Images to Predictions

End-to-End Pipeline Overview



Processing time: ~ 20 min/organoid
(dominated by segmentation)

Compression: 2 GB $\rightarrow$ 10 MB
(1000 $\times$ reduction)

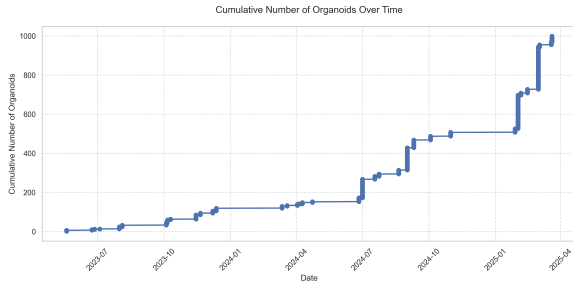
Collaborative Dataset

ANR Morpheus Project

- 22 months of data collection
- May 2023 – February 2025
- IPMC Nice + Paris Cité

Dataset Statistics:

- 1,311 imaged samples
- 2,272 extracted organoids
- **500 well-differentiated** selected
- ~250 per class



Faster Cellpose: Segmentation Optimization

Problem: Cellpose is slow

- $30 \text{ sec/slice} \times 200 \text{ slices} \times 1000 \text{ organoids} = \mathbf{2,500 \text{ hours}}$

Our Solution: Faster Cellpose

- ① **Knowledge Distillation:** Smaller student network (−50% params)
- ② **Weight Pruning:** Remove 30% low-magnitude weights
- ③ **Inference Optimization:** Larger batches, mixed precision

Result

5× speedup while preserving $F1 = 0.95$

2,500 hours → 500 hours

Segmentation Methods Comparison

Method	F1 Score	Time/Slice	Speed-up
Geometric (ellipses)	0.88	3 sec	10×
Faster Cellpose	0.95	6 sec	5×
Original Cellpose	0.98	30 sec	1×

Choice: Faster Cellpose = optimal trade-off

- Segmentation quality → Graph quality → GNN performance

Geometric Graph Construction

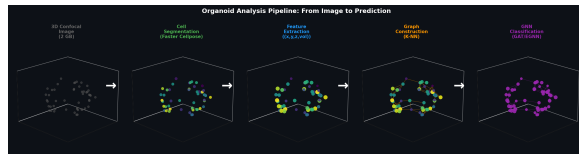
Node Features (4D):

- 3D centroid: (x, y, z)
- Cell volume
- Normalized (centered, scaled)

Edge Construction:

- K-nearest neighbors ($K = 10$)
- Euclidean distance
- Symmetrized + radial cutoff

▶ Animation: `graph_construction.gif`



From segmentation to graph

Synthetic Generation via Point Processes

Key Insight: Cell distribution differs

Cystic — Poisson

Homogeneous, spatially random

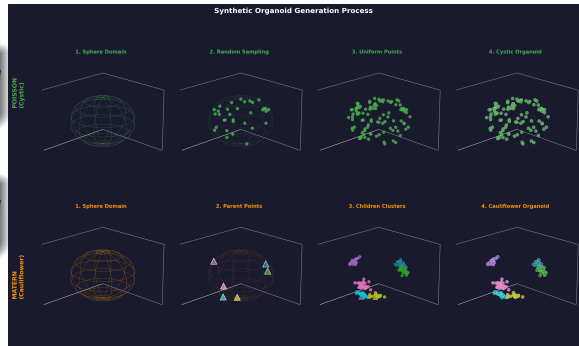
Cauliflower — Matérn

Parent points + clustered children

Generated: 100,000 synthetic organoids

Perfect labels, infinite supply

▶ `point_processes.gif`



Generation pipeline steps

Transfer Learning: Synthetic $\rightarrow$ Real

Phase 1: Pre-training on 70,000 synthetic organoids

- Task: Regress Matérn parent number
- Learn spatial organization patterns

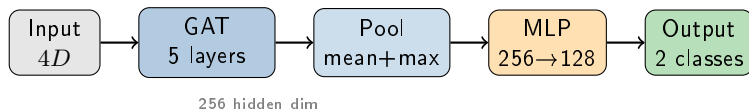
Phase 2: Fine-tuning on 500 real organoids

- Reinitialize classification head
- Reduced learning rate

Benefits

- **4 $\times$ data efficiency:** 125 pre-trained $\approx$ 500 from-scratch
- **3 $\times$ faster convergence**
- **+8% accuracy improvement**

Final Architecture Configuration



Training

Optimizer	AdamW
LR (pre-train)	10^{-3}
LR (fine-tune)	10^{-4}
Dropout	0.15

Compute

Pre-training	200 ep, 48h
Fine-tuning	100 ep, 30 min
Hardware	RTX 3080

- ① **Complete automated pipeline** from raw images to predictions
- ② **Faster Cellpose**: $5\times$ speedup, $F1 = 0.95$
- ③ **Synthetic generation**: 100,000 organoids with perfect labels
- ④ **Transfer learning**: $4\times$ reduction in annotation needs

Part III

Experimental Results

Validation of Our Approach

Research Question:

When should we prefer GNNs over classical methods (Ripley's K)?

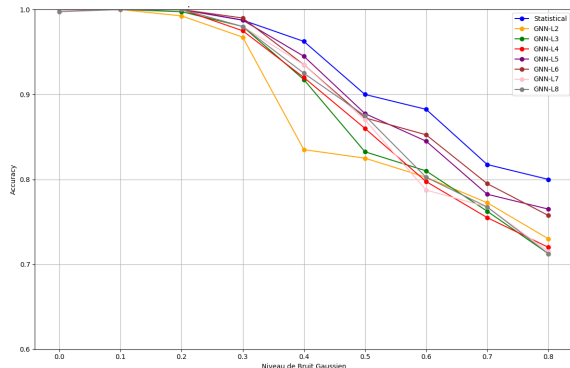
Protocol:

- Synthetic data with controlled ground truth
- Spherical point distributions: Poisson vs Matérn
- Two noise types: Gaussian (position) + Salt-and-pepper (points)

Comparison:

- GNN classifiers (various depths)
- Random Forest on Ripley's K, F, G functions

Result 1: Noise Robustness



Observation:

Spatial statistics are **more robust to noise**

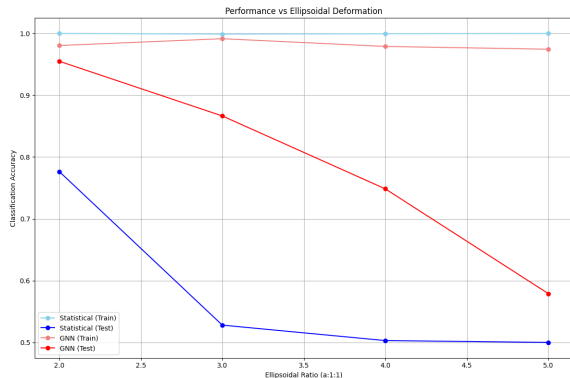
Why?

- Ripley's functions average over many point pairs
- Natural smoothing of noise
- GNNs propagate noise through layers

Optimal GNN depth: 5-6 layers

- Deeper → over-smoothing

Result 2: Geometric Generalization



Training: Spherical distributions

Testing: Ellipsoids (ratio 1:1 $\rightarrow$ 5:1)

Results:

- Spatial stats: 100% $\rightarrow$ 65% (-35%)
- GNNs: 95% $\rightarrow$ 82% (-13%)

Why?

- Ripley's K assumes isotropy
- GNNs learn topological patterns

Conclusion

GNNs excel when morphologies vary — exactly our situation!

Lesson from Comparative Study

GNNs Excel When:

- Morphologies vary
- Shapes are irregular
- Geometry is non-spherical

⇒ **Real organoids**

Spatial Statistics Excel When:

- Geometry is perfectly spherical
- Consistent across samples
- Low noise

⇒ **Idealized conditions**

Foundation for Methodological Choices

For cystic vs cauliflower with variable morphologies, GNNs are appropriate

Performance on Synthetic Data

Task: Regression of Matérn parent number

Test set: 15,000 synthetic organoids

Architecture	MSE	Improvement vs GCN
GCN (baseline)	0.198	—
DeepSets	0.145	+27%
EGNN	0.137	+31%
GAT	0.118	+40%

GAT wins: Attention enables adaptive neighbor weighting

Synthetic Results Analysis

Why GAT wins:

- Attention learns which neighbors matter most
- Adaptive to clustering patterns

DeepSets' good performance:

- Global statistics capture substantial information
- But local structure still helps (GAT beats DeepSets)

EGNN trade-off:

- Slightly behind GAT in raw numbers
- But guaranteed equivariance
- Equivariance divides MSE by $2.8\times$

Dataset: 500 well-differentiated organoids

Evaluation: 5-fold cross-validation

84% Accuracy
(GAT + pre-training)

Cauliflower:

- Precision: 93%
- Recall: 74%

Cystic:

- Precision: 78%
- Recall: 95%

Confusion Matrix

	Pred. Cauliflower	Pred. Cystic	
Cystic Cauliflower	28	10	63/75 correct = 84% accuracy
	2	35	

Main confusion: Cauliflower $\rightarrow$ Cystic

- Low-deformation cauliflowers appear quasi-spherical
- Possible intermediate/transitional phenotypes

Training Strategy	Accuracy
GAT from scratch (100% data)	76%
GAT pre-trained + fine-tuned	84%
<i>Improvement</i>	+8%

Data Efficiency:

- Pre-trained with 25% data (125 organoids): 78%
- From-scratch with 100% data: 76%

4× Reduction in Annotation Requirements

Transformative for practical applications

Learning Curves: Transfer Learning Advantage

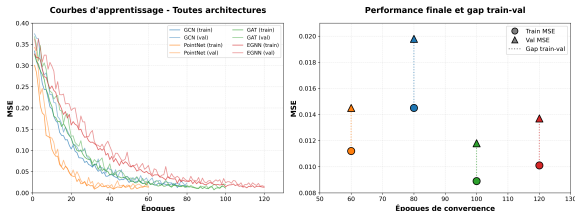
Data efficiency gains:

Data	Gain
10%	+13%
25%	+11%
50%	+10%
100%	+8%

Convergence speed:

- Pre-trained: 20-30 epochs
- From-scratch: 80-100 epochs

⇒ 3× faster



Training convergence comparison

Inference:

- **200+ organoids/minute** (GPU batch)
- Enables high-throughput screening

Memory:

- ~8 GB GPU memory
- Compatible with standard hardware

Reproducibility:

- Identical outputs every time
- No inter-observer variability

Scalability:

- 1000 organoids: ~17 hours
- 20 GPUs in parallel

Experimental Results: Summary

- ① **GNNs > spatial statistics** for variable morphologies
- ② **GAT achieves best performance:**
 - MSE 0.118 (synthetic)
 - 84% accuracy (real)
- ③ **Transfer learning:**
 - 4× data efficiency
 - 3× faster convergence
- ④ **Practical pipeline:**
 - 200+ organoids/minute
 - Fully automated, reproducible

Identified Limitations

- **Segmentation dependency:**
 - Errors propagate through pipeline
- **Validation scope:**
 - Prostate organoids only
 - Generalization to brain, liver, etc. needs testing
- **No inter-laboratory validation:**
 - Different microscopes, protocols
- **No interpretability analysis:**
 - Which cells/regions drive predictions?

Part IV

Conclusion and Perspectives

Contribution 1: Automated Pipeline

Complete End-to-End Pipeline

From raw 3D images to phenotype predictions

Features:

- Full workflow: preprocessing → segmentation → classification
- **1000× compression** while preserving biological information
- **Open-source** codebase available to community

Contribution 2: Segmentation Optimization

Faster Cellpose

5× speedup through knowledge distillation and pruning

Maintains $F1 = 0.95$

Additional contribution:

- Geometric method (ellipse detection): 15× speedup
- Useful for ultra-rapid primary screening

Impact: Makes high-throughput analysis practical

Contribution 3: Geometric Graphs and GNNs

Graph-Based Representation

Geometric graphs capture cellular relational structure

Contributions:

- Systematic comparison: GAT, DeepSets, EGNN, GCN
- Demonstrated value of **$E(3)$ -equivariant architectures**
- **GAT**: best accuracy
- **EGNN**: guaranteed geometric robustness

Contribution 4: Synthetic Data Generation

Point Process-Based Generation

Poisson + Matérn processes for synthetic organoids

Results:

- **100,000 synthetic organoids** with perfect labels
- **+8% accuracy** improvement
- **4× reduction** in annotation needs

Generalizable: Applicable to other spherical/ellipsoidal structures

Published at GRETSI 2025

Rigorous comparison: GNNs vs Spatial Statistics

Findings:

- GNNs: Better **geometric generalization**
- Spatial statistics: Better **noise robustness**

Impact: Guides method selection based on data characteristics

Methodological Extensions:

- Multi-scale graphs (cell $\rightarrow$ region $\rightarrow$ organoid)
- Graph Transformers for global attention
- Alpha-shape morphological descriptors

Multi-Modal Integration:

- Spatial transcriptomics + imaging
- Temporal data from time-lapse imaging

Clinical Validation:

- Prospective studies on patient cohorts
- Therapeutic response prediction

Therapeutic Response Prediction:

- Patient-derived organoids
- Test multiple treatments in silico
- Predict best therapy for individual patients

Generative Graph Models:

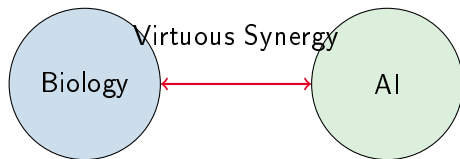
- In silico organoid design
- Optimize culture protocols computationally

Societal Impact:

- 3Rs principles: Replace, Reduce, Refine animal experimentation
- Accelerate transition to organoid-based screening

Geometric Graph Neural Networks

are a powerful tool for 3D organoid analysis



Open-source code available for the community

Thank you for your attention.
I am happy to take your questions.

Backup Slides

Anticipated Questions

① Why GAT over EGNN?

Trade-off: GAT wins on accuracy; EGNN wins on theoretical guarantees

② Synthetic data validation?

Empirical gains (+8%) demonstrate practical utility

③ Generalization to other organoids?

Principles transferable; requires fine-tuning Cellpose

④ Why only 500 organoids?

Quality over quantity — ensuring label-morphology consistency

⑤ Clinical deployment?

Would require regulatory validation (IVDR/FDA), multi-site testing

Timing Checkpoints

Section	Target Time
After Introduction (Slide 8)	~7 min
After Foundations (Slide 22)	~20 min
After Methodology (Slide 32)	~29 min
After Results (Slide 46)	~40 min
End	~47 min

If running long: Abbreviate slides 38-39 (synthetic results)

If running short: Expand perspectives (slides 53-54)